Exam 2

98

1) SFRs have the following inherent safety features:

- Sodium has fantastic thermal properties, such as high conductivity for emergency heat removal, high heat capacity, and high margin to boiling

6/6

- Negative temperature reactivity; when temperature increases, the density of core materials decreases, making them more neutron transparent and increasing leakage, which decreases reactivity. This negative feedback loop prevents runaway reactions, even without operator intervention.

2) Alloying (especially with Zr) tends to decrease the amount of swelling. It also widens the temperature range of the γ BCC phase, raising melting point and partially avoiding the anisotropically swelling α phase. Zr can also form a protective rind that reduces FCCI.

6/6

3) Low smear density gives the fuel pin a spatial margin from the cladding into which it can swell. Large amounts of swelling are unavoidable in metal fuel. W/o the low SD, the swelling would rupture cladding through FCMI. The large gas plenum exists to mitigate the pressure rise in the fuel element from gaseous fission products.

9/10

- needed a little more

→ Zr is good for FCCI in general.

4) In U-Zr systems, there are 3 "zones" with different Zr concentrations:

14/14

- Zr-rich zone in the radial center. This zone is hot enough to be in γ-U phase, which readily dissolves high concentrations of Zr.

- Zr-depleted zone outside the center zone. This zone has a conglomeration of U phases that do not have high solubility for Zr (e.g. β-phase).

- Slightly Zr-rich outer zone. This zone is cool enough to support a combination of α-U and $\delta(UZr_2)$ phases.

Above are thermodynamic drivers. The actual diffusion mechanism is related to Soret diffusion, which suggests that Zr would move up a thermal gradient and U would move down. This could also explain the Zr-rich center zone. Soret

The main impact of this phase/constituent distribution is anisotropic swelling. The outer, stronger α/δ phase acts as a pressure vessel for the very porous and swelling inner γ phase. This causes the radial stress component to be double the axial component, causing a mismatch (anisotropy) in strain rates radially/axially. Another impact of the redistribution is that the Zr-rich outer zone may form a protective rind if oxygen impurities are present (ZrO_2). Also, FP lanthanides tend to move toward the cladding, leading to FCCI issues.

of the fuel pin overall

8/8

5) At low burnups, thermal conductivity degrades with burnup as voids form in the fuel from gaseous fission products. However, at some burnup, the voids interconnect and pressure is relieved to the gas plenum. This allows bond sodium to infiltrate the fuel, creating a sudden and large increase in thermal conductivity.

14/14

6) FCCI is Fuel-Cladding Chemical Interaction. FCCI can mechanically weaken the cladding and create the potential for low-melting point phases to form in the fuel element (particularly $(U,Pu)_6Fe$). The primary fuel species participating are fission product lanthanides, which diffuse into the cladding. The primary cladding participant is Fe, which diffuses into fuel and could create phases with a low eutectic melting point. FCCI particularly occurs where there are lanthanide deposits along the cladding in a α/δ U-Zr crack.

Ce, Nd

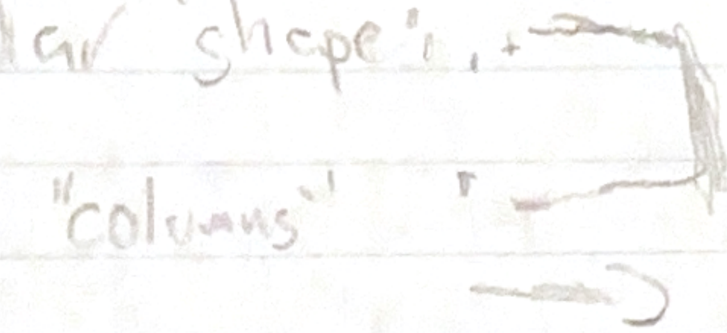
7) Restructuring in Mox fel creates four layers: (inside to outside)

12/12

- Central void; forms as pores within the fuel move up the thermal gradient and gather
- Columnar grain region; T is high enough that pores traveling to the center leave behind grains that are preferentially oriented in the direction pores move (along thermal gradient).
- Equiaxed region: T is high enough to increase grain size, but not high enough to form columnar grains.

- As-sintered region; grain size same as fabricated.

Pores move up the thermal gradient because constituents on the hotter side of the pore evaporate and then condense on the cool side, causing net motion of the pore. Pores also develop a lenticular shape:



This leaves behind the "columns"

9/10

8) Pu is most concentrated in the center (leading edge of central void) because Pu is less volatile, which means it does not move down the thermal gradient as much by the mechanism of question 7. O is most concentrated at the periphery due to preferential movement of volatile species down the thermal gradient.

- somewhat

12/12

9) JOG is joint oxide gain. It refers to the buildup of volatile FP oxides on an outer layer between fuel and cladding. This joint oxide layer acts as a buffer for FCMI. Postive impact However, certain elements (especially Cs) can chemically attack the cladding in the presence of high oxygen potential. This FCCI is a negative impact. Using hypostoichiometric fuel can reduce oxygen potential at cladding interface.

- Rth effects

10) Sodium corrodes steels in two ways:

8/8

- Dissolution of steel components (especially Ni) in liquid Na. This can turn austenitic steels ferritic. Steel components dissolve at high T (hot) and deposit at low T (THX), creating a thermal gradient mass transfer effect. Oxygen impurity in Na can increase the effective solubility of cladding components.

- Chemical interactions between cladding and Na impurities. Of particular interest is the interaction between Cr in steel and C impurities. High Cr steels will tend to carbeurize, while low Cr steels tend to decarbeurize.

Any SG leak could be catastrophic, as Na reacts violently (exothermically) with water. This is a big SFR drawback.